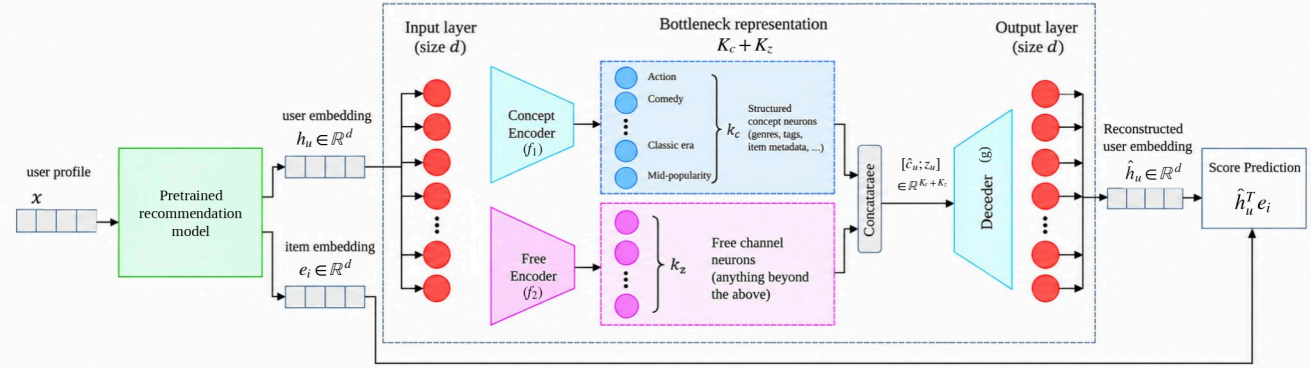

user profile
x
Pretrained recommendation model
user embedding
$h_u \in \mathbb{R}^d$
item embedding
$e_i \in \mathbb{R}^d$
Input layer (size d)
Concept Encoder (f_1)
Free Encoder (f_2)
Bottleneck representation
$K_c + K_z$
Action
Comedy
Classic era
Mid-popularity
k_c
Structured concept neurons (genres, tags, item metadata, ...)
k_z
Free channel neurons (anything beyond the above)
Concatcatee
$[\hat{c}_u; z_u]$
$\in \mathbb{R}^{K_c + K_z}$
Deceder (g)
Output layer (size d)
Reconstructed user embedding
$\hat{h}_u \in \mathbb{R}^d$
Score Prediction
$\hat{h}_u^T e_i$